Investigating Robustness of Decentralized Swarm Systems Under Robot Failure and Sensing Noise

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Abstract

Decentralized swarm systems are capable of achieving complex collective behavior through local interactions without centralized control. Understanding the robustness of these systems under uncertainty is important for applications in environments where communication is limited and individual robot failures are expected. This study investigates the effects of robot failures and sensing noise on the performance of a decentralized, pheromone-based swarm simulation developed in Python. A two-dimensional foraging environment containing 50 autonomous robots and 100 randomly distributed resources was used to evaluate swarm behavior. Two independent sources of uncertainty were examined: per-tick robot failure probability and Gaussian sensing noise. A total of 140 simulation trials were conducted, and swarm performance was evaluated using completion percentage, simulation time, and resource collection rate. The results showed that increasing robot failure probability substantially reduced both completion percentage and swarm lifetime, while sensing noise primarily affected collection efficiency. Interestingly, moderate sensing noise produced the shortest average completion times and highest collection rates, suggesting that limited sensing uncertainty may encourage more effective exploration. These findings demonstrate that decentralized pheromone-based swarms are resilient to moderate sensing uncertainty but remain vulnerable to persistent robot failures. The study provides insight into the robustness of decentralized multi-agent systems and highlights directions for future swarm robotics research.

1. Introduction

Swarm robotics is a field of robotics inspired by the collective behavior of biological systems such as ant colonies, bee swarms, and flocks of birds (Bonabeau et al., 1999). Rather than relying on a centralized controller, swarm systems consist of many relatively simple agents that interact only with their local environment and neighboring agents. Through these decentralized interactions, complex group behaviors can emerge that allow the swarm to accomplish tasks that would be difficult or impossible for a single robot.

One of the primary advantages of decentralized swarm systems is their potential robustness. Because decision-making is distributed among many independent agents, the failure of individual robots does not necessarily prevent the swarm from completing its objective. This property makes swarm robotics particularly attractive for applications in hazardous or inaccessible environments, including disaster response, environmental monitoring, planetary exploration, and search-and-rescue missions, where communication may be limited, and individual robot failures are expected.

Despite these advantages, decentralized swarms remain vulnerable to uncertainty. Robots may experience sensor inaccuracies, navigation errors, or permanent failures that disrupt the collective behavior of the system. Understanding how these sources of uncertainty affect swarm performance is essential for designing reliable multi-agent systems capable of operating in real-world environments.

This study investigates the robustness of a decentralized swarm through a custom-built foraging simulation based on pheromone-inspired communication. Robots search for randomly distributed resources, transport them to a central collection site, and deposit virtual pheromone trails that guide future robots toward successful paths. Two sources of uncertainty were examined independently: per-tick robot failure probability and sensing noise introduced through Gaussian error. Performance was evaluated over 140 simulation trials using metrics including completion percentage, simulation time, and resource collection rate.

The objective of this research is to determine how decentralized pheromone-based swarms respond to increasing levels of robot failure and sensing uncertainty, and to identify the conditions under which collective behavior remains robust despite these disruptions.

2. Background

2.1 Swarm Intelligence

Swarm intelligence is a form of collective behavior in which many simple agents interact locally to produce coordinated global behavior. Unlike traditional robotic systems that rely on centralized control, swarm systems achieve complex objectives through decentralized decision-making. Each agent follows a relatively small set of behavioral rules, yet the interactions among many agents allow the swarm to solve problems such as foraging, path planning, and exploration.

Natural systems provide many examples of swarm intelligence. Ant colonies collectively locate food sources, birds coordinate flocking behavior, and schools of fish respond rapidly to predators without any individual directing the group. These biological systems demonstrate that sophisticated collective behavior can emerge from simple local interactions.

2.2 Decentralized Control

In decentralized swarm systems, no robot possesses complete knowledge of the environment or directs the actions of other agents. Instead, each robot makes decisions based solely on local information obtained through sensors and interactions with its surroundings. This decentralized approach offers several advantages, including scalability, fault tolerance, and flexibility (Brambilla et al., 2013). As additional robots are introduced, the swarm can continue operating without requiring changes to a central controller, and the failure of individual robots is less likely to cause complete system failure.

However, decentralized systems also present unique challenges. Because no single agent coordinates the swarm, overall performance depends on the reliability of local interactions. Sensor inaccuracies, communication limitations, and robot failures can all influence the collective behavior that emerges from these interactions.

2.3 Pheromone-Based Communication

Many swarm algorithms are inspired by the foraging behavior of ants, which communicate indirectly through a process known as stigmergy. Rather than communicating directly with one another, ants deposit pheromone trails while returning to their nest with food. Other ants detect these trails and are more likely to follow paths that have previously led to successful resource discovery (Theraulaz & Bonabeau, 1999). As pheromones gradually evaporate, outdated paths become less attractive, allowing the colony to adapt to changes in its environment.

The simulation developed in this study adopts this same principle. Robots deposit virtual pheromone markers while transporting collected resources back to a central base. These markers guide subsequent robots toward productive search regions, allowing efficient resource collection to emerge without centralized coordination or direct communication between robots.

3. Methodology

3.1 Simulation Environment

A two-dimensional swarm simulation was developed in Python using the Pygame library to investigate the robustness of decentralized swarm systems under varying levels of robot failure and sensing noise. Figure 1 displays the simulation environment, which consists of a rectangular arena containing a central collection base, a swarm of autonomous robots, and randomly distributed resources.

At the start of each simulation, robots and resources were assigned random positions within the environment. The swarm contained 50 robots, while 100 resources were placed throughout the arena. Random initial conditions were regenerated for every trial to ensure that the experimental results reflected average swarm behavior rather than a single environmental configuration.

For the baseline and noise experiments, simulations terminated when all resources had been collected. For failure experiments, simulations could also terminate if all robots had permanently failed before collecting every resource.

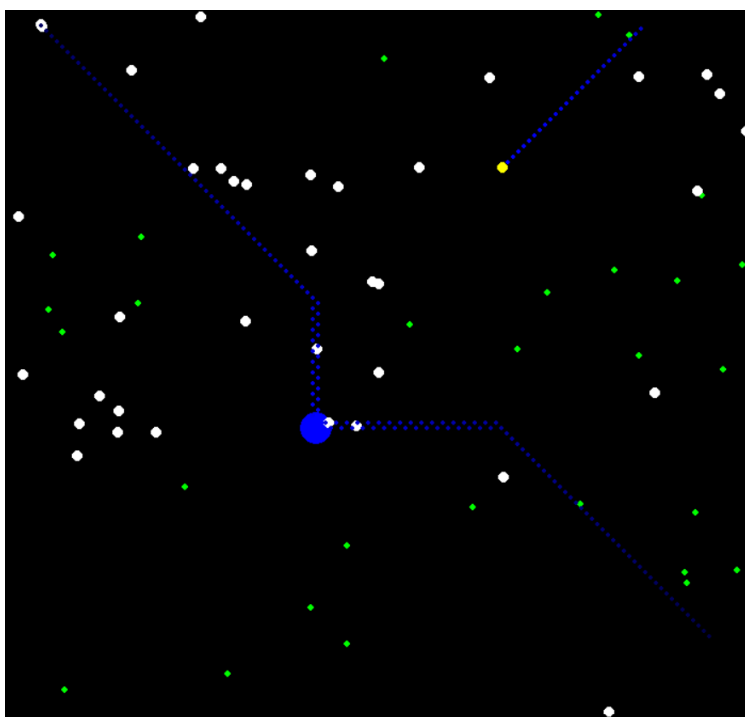

Figure 1. Simulation environment showing searching robots (white), returning robots (yellow), resources (green), the central collection zone (blue circle), and pheromone trails (small blue dots).

3.2 Robot Behavior

Each robot operated independently according to a decentralized control algorithm. Robots existed in one of two behavioral states: search mode and return mode.

During search mode, robots performed a random walk while scanning their surroundings for nearby resources or existing pheromone trails. If a resource was detected within the sensing radius, the robot navigated toward it. Upon reaching the resource, it collected the object and transitioned into return mode.

While returning to the collection base, robots deposit virtual pheromone markers at their current positions. These pheromone trails served as indirect communication for the remainder of the swarm by indicating previously successful paths to resource locations. Once a robot reached the collection base, the resource was deposited, the carrying state was cleared, and the robot resumed searching for additional resources.

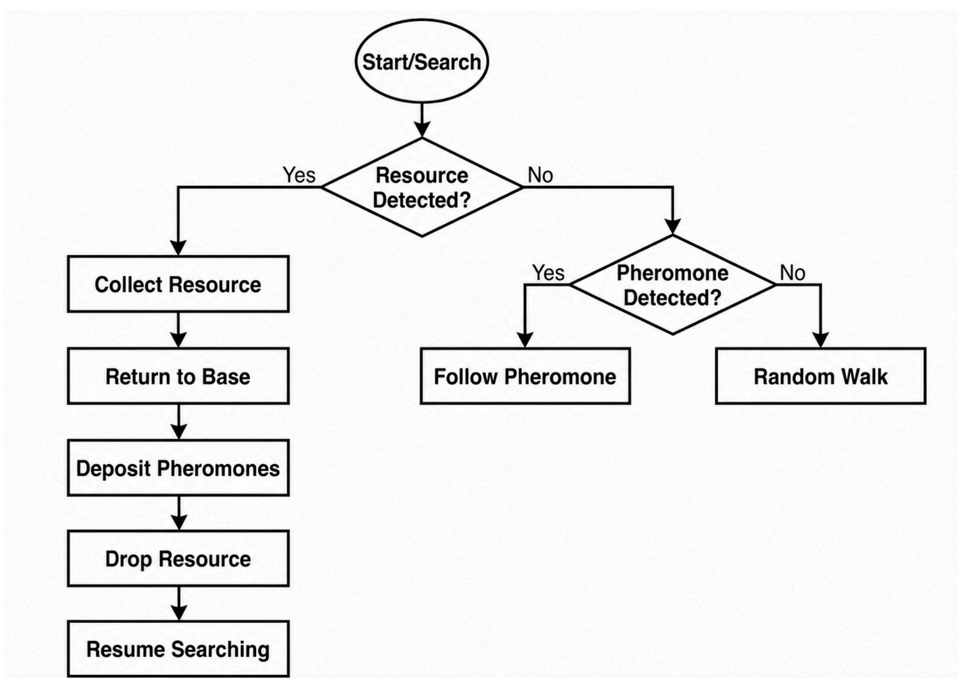

Figure 2. Robot behavior flowchart

3.3 Pheromone Model

The simulation implemented a simplified pheromone-based communication system inspired by a colony's foraging behaviors. Robots carrying resources continuously deposited pheromone markers along their return path to the collection base.

Searching robots were capable of detecting nearby pheromone markers and preferentially moving toward them when no resources were directly visible. Each pheromone marker gradually evaporated over time, with its intensity decreasing every simulation tick. Once its intensity reached zero, the marker was removed from the environment. This evaporation mechanism allowed outdated paths to disappear naturally while reinforcing frequently traveled routes.

3.4 Failure and Noise Models

Two independent sources of uncertainty were investigated.

The first was per-tick robot failure probability. During every simulation tick, each active robot had a specified probability of permanently failing. Failed robots immediately ceased all movement and no longer participated in resource collection or pheromone deposition. Three failure probabilities were evaluated in addition to the baseline condition (0.005%, 0.02%, and 0.05%). Very small per-tick failure probabilities were used because failure events were evaluated every simulation tick; consequently, the cumulative probability of a single robot surviving decreases according to $P\_survival = (1 - p)^n$, where p denotes the per-tick failure probability, $0 \leq p \leq 1$, and n denotes the number of elapsed simulation ticks, $n \in \mathbb{N}$.

The second source of uncertainty was sensing noise. Gaussian noise was added independently to the perceived x- and y- coordinates of detected resources before navigation decisions were made. This caused robots to estimate resource locations with varying degrees of error while leaving all other aspects of the simulation unchanged. Three noise standard deviations (20px, 50px, and 100px) were investigated.

Failure and sensing noise were studied independently to isolate the effect of each variable on swarm performance.

3.5 Experimental Design

The study consisted of seven experimental conditions: one baseline configuration, three failure-only conditions, and three noise-only conditions. Each condition was repeated for 20 independent trials using newly randomized initial robot and resource positions.

Three performance metrics were recorded for every trial:

- Completion percentage, representing the proportion of resources successfully collected.
- Simulation time, representing the total simulated time required for task completion or swarm collapse.
- Collection rate, defined as the average number of resources collected per unit of simulated time.

To promote reproducibility, the complete simulation source code, experimental datasets, and supporting documentation are publicly available in the accompanying GitHub repository: https://github.com/AmbitiousElectricallion/Swarm-Simulation

4. Results

4.1 Baseline Performance

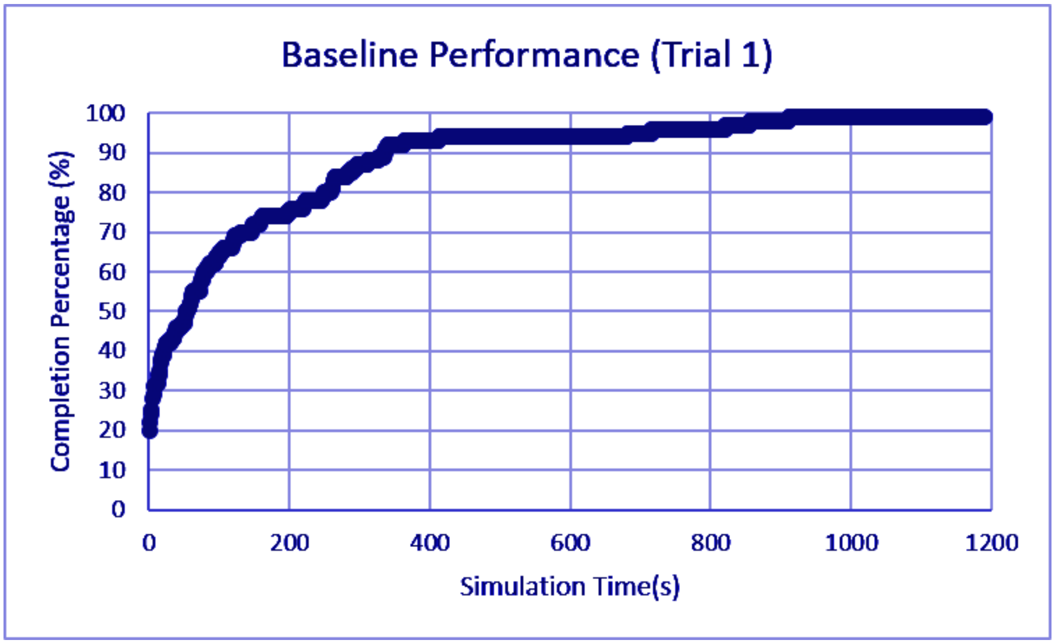

Figure 3. Completion percentage over time during a representative baseline trial.

Figure 3 illustrates the progression of resource collection during a representative baseline trial. Resource collection initially increased rapidly as multiple robots simultaneously discovered and transported nearby resources. However, after approximately 400 seconds, the rate of resource collection gradually decreased. As fewer resources remained in the environment, robots spent more time searching for the remaining scattered resources before returning them to the collection base.

Across all 20 baseline trials, the mean completion time was 1186.84 s (SD = 397.63s), corresponding to a mean collection rate of 0.0937 resources/s (SD = 0.0298 resources/s). Although completion times varied because of randomized initial conditions, every baseline trial inevitably completed the task due to simulation termination logic.

4.2 Effect of Robot Failure Probability

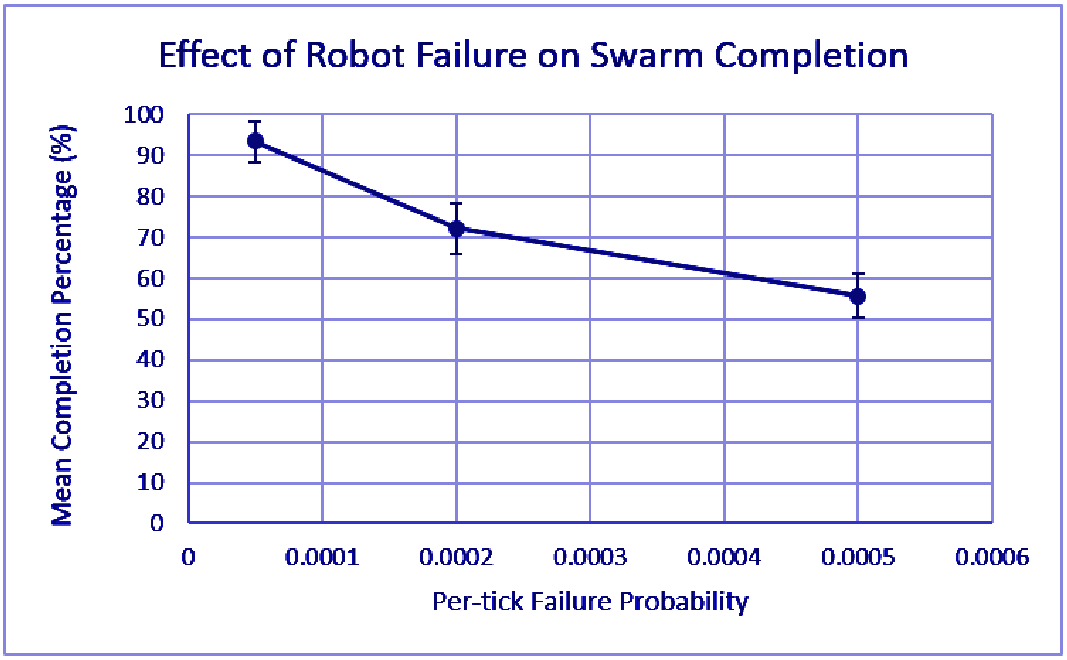

Figure 4. Mean completion percentage versus per-tick failure probability.

Figure 4 shows the relationship between per-tick failure probability and mean completion percentage. Increasing the failure probability produced a clear decline in task completion. Mean completion percentage decreased from 100% in the baseline condition to 93.4%, 72.2%, and 55.7% for failure probabilities of 0.005%, 0.02%, and 0.05%, respectively.

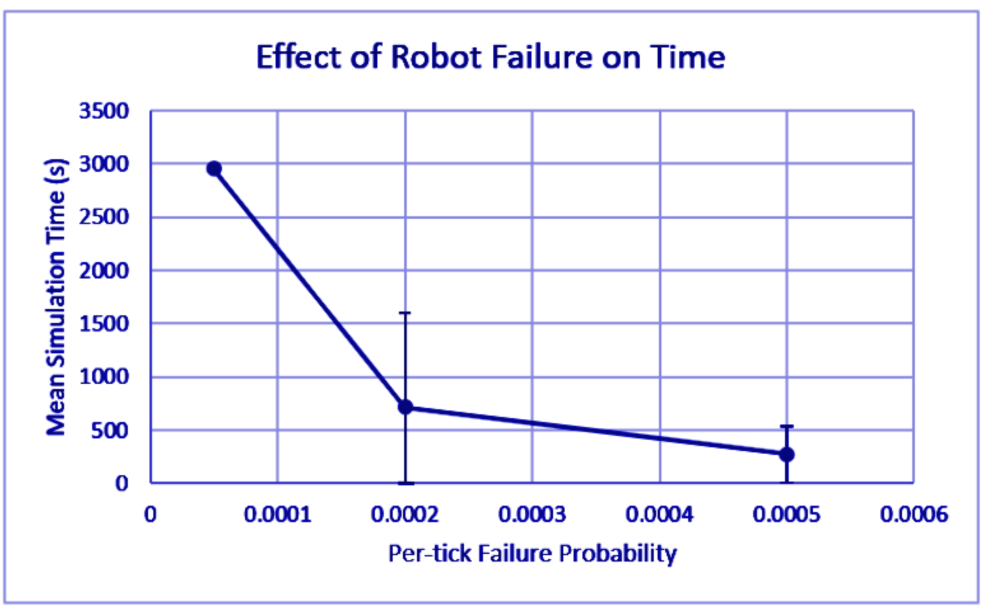

Figure 5. Mean simulation time versus per-tick failure probability.

Figure 5 presents the corresponding mean simulation times. At the lowest failure probability (0.00005), the swarm remained active for an average of 2954.11 s (SD = 887.21 s) before the simulation ended. As the failure probability increased to 0.02% and 0.05%, mean simulation time decreased to 718.35 s (SD = 267.61s) and 274.67 s (SD = 67.58 s), respectively. These reductions occurred because robots failed more rapidly, causing the swarm to lose active agents before completing the task.\

4.3 Effect of Sensing Noise

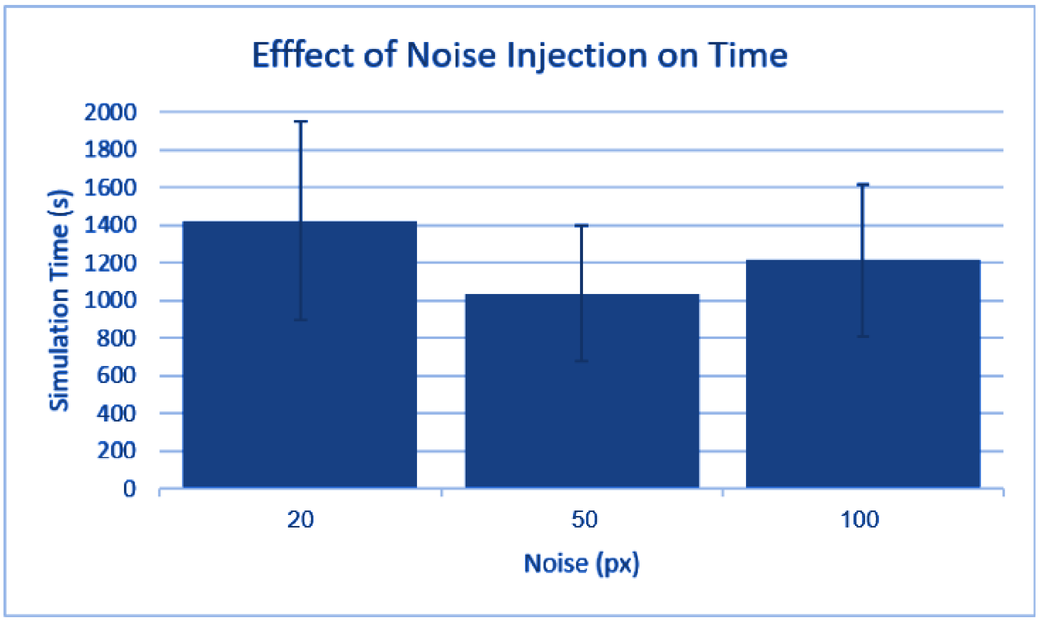

Figure 6. Mean simulation time for different sensing noise standard deviations.

Completion percentage was not used to compare the noise experiments because the simulation terminated once all 100 resources had been collected. As a result, every noise condition reached 100% completion, making completion percentage unsuitable for distinguishing performance differences. Instead, the effects of sensing noise were evaluated using simulation time and collection rate, which more effectively captured differences in swarm efficiency.

Figure 6 compares the mean simulation time for each sensing noise level. The 50 px condition produced the shortest mean completion time (1036.77 s, SD = 360.41 s), compared with 1423.03 s (SD = 529.56 s) for 20 px and 1213.40 s (SD = 404.91 s) for 100px.

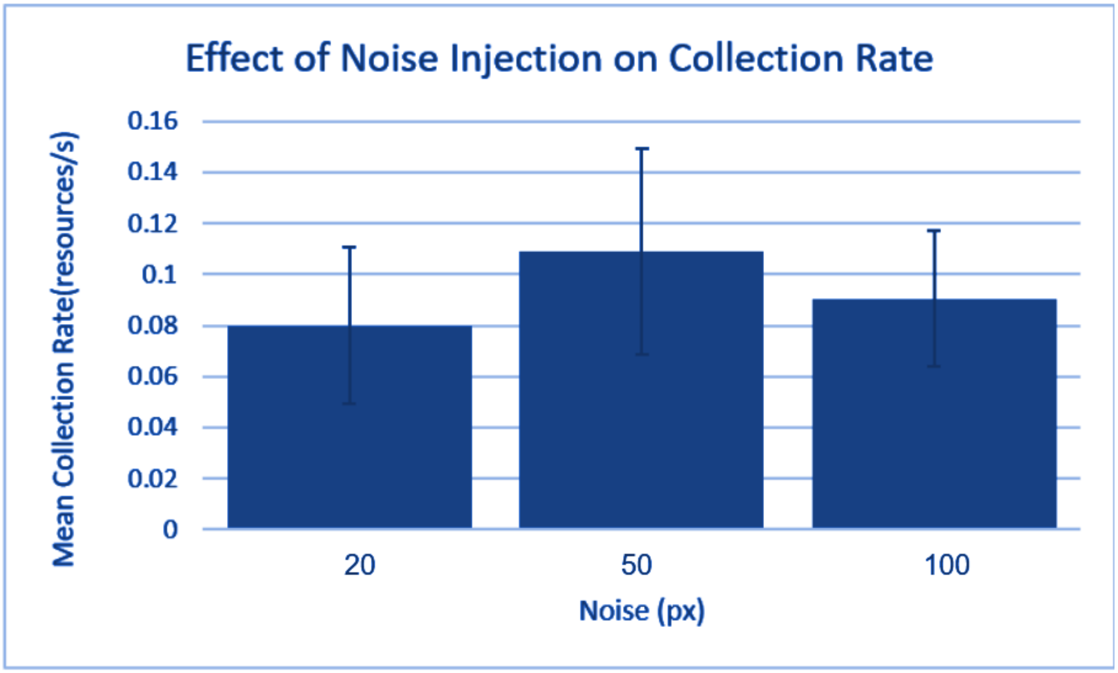

Figure 7. Mean collection rate for different sensing noise standard deviations.

Figure 7 presents the corresponding mean collection rates. The 50 px condition achieved the highest mean collection rate (0.1090 resource/s), followed by 100px (0.0904 resources/s) and 20 px (0.800 resources/s). These results indicate that moderate sensing noise improved collection efficiency under the conditions investigated.

5. Discussion

The results demonstrate that decentralized swarm systems exhibit different levels of robustness depending on the type of uncertainty introduced. Permanent robot failures had a significantly greater impact on overall performance than sensing noise, indicating that the swarm was more resilient to imperfect information than to the irreversible loss of participating agents.

As the per-tick failure probability increased, both the completion percentage and the simulation time decreased substantially. Because failed robots permanently ceased movement and no longer deposited pheromone trails, the swarm gradually lost its ability to maintain efficient exploration and resource transport. At higher failure probabilities, simulations frequently terminated because all robots had failed before completing the task. These results suggest that maintaining a sufficient number of active agents is critical for sustaining decentralized collective behavior.

In contrast, sensing noise did not prevent successful task completion under the conditions investigated. Although robots navigated using imperfect estimates of resource locations, every noise condition ultimately collected all available resources. Instead, sensing noise primarily influenced the efficiency of resource collection. Interestingly, the moderate noise condition (50 px) produced both the shortest average simulation time and the highest average collection rate.

One possible explanation is that moderate sensing uncertainty increased exploration throughout the environment. Without noise, robots may have been more likely to repeatedly search similar regions or follow identical paths. A moderate amount of randomness encouraged broader coverage of the search space while remaining sufficiently accurate for successful resource collection. However, when sensing noise became too large, navigation accuracy declined enough to offset any exploratory benefit, resulting in longer completion times and lower collection rates.

Although the simulation incorporated multiple sources of randomness, including robot starting positions, resource placement, and random exploration behavior, these factors were intentionally regenerated for every trial. Repeating each experimental condition twenty times allowed performance to be evaluated using mean values and standard deviations rather than relying on individual simulations. This experimental design reduced the influence of random variation while preserving the stochastic nature of decentralized swarm systems.

Overall, the findings suggest that pheromone-based decentralized swarms are capable of maintaining effective collective behavior under moderate sensing uncertainty but remain considerably more vulnerable to persistent robot failures. These observations support the importance of robustness analysis when designing decentralized multi-agent systems intended for real-world operation.

6. Limitations

Some limitations should be considered when interpreting the results of this study. First, the simulation represents a simplified environment without obstacles, terrain variation, or dynamic changes. Real-world swarm robotic systems often operate in significantly more complex environments that may influence navigation and communication.

Second, all robots were assumed to be identical in their sensing capabilities, movement speed, and behavioral rules. In practical applications, differences in hardware performance, battery life, or sensor quality may introduce additional variability not captured by this model.

The failure model also represents an idealized scenario in which robots permanently fail according to a fixed per-tick probability. Real robotic failures are typically influenced by factors such as battery depletion, mechanical wear, environmental hazards, or communication loss. Similarly, sensing noise was modeled using Gaussian error applied only to perceived resource locations, whereas real sensors may experience more complex forms of uncertainty.

Finally, although multiple sources of randomness were present throughout the simulation, including robot initialization, resource placement, and random exploration, each experimental condition was repeated across twenty independent trials. Mean values and standard deviations were used to reduce the influence of stochastic variation while preserving the decentralized nature of the system.

7. Future Work

Several extensions could improve both the realism and scope of this simulation. Future work could introduce static or dynamic obstacles to investigate how decentralized swarms adapt to more complex environments. Additional failure models, such as battery depletion, temporary communication loss, or recoverable robot failures, would provide a more realistic representation of field conditions.

The pheromone model could also be expanded by allowing robots to deposit pheromone trails with varying strengths or adapt evaporation rates based on environmental conditions. Investigating heterogeneous swarms of robots with varying sensing capabilities or movement speeds would further improve the simulation's realism.

Finally, future studies could examine the combined effects of sensing uncertainty and robot failures. While these variables were investigated independently in this study to isolate their individual impacts on swarm performance, understanding their interaction would provide additional insight into the robustness of decentralized multi-agent systems operating under multiple simultaneous sources of uncertainty.

8. Conclusion

This study investigated the robustness of a decentralized, pheromone-based swarm system under two independent sources of uncertainty: per-tick robot failure probability and sensing noise. A custom two-dimensional swarm simulation was developed to evaluate how these factors influenced resource collection performance using metrics including completion percentage, simulation time, and collection rate.

The experimental results showed that increasing robot failure probability significantly reduced swarm performance. Higher failure probabilities lowered completion percentages and caused simulations to terminate earlier as active robots were permanently removed from the system. These findings demonstrate that maintaining a sufficient population of functioning agents is essential for sustaining decentralized collective behavior.

In contrast, sensing noise primarily affected the efficiency of resource collection rather than the swarm's ability to complete the task. Moderate sensing noise produced the shortest average simulation times and the highest collection rates, suggesting that a limited amount of uncertainty may encourage more effective exploration without substantially reducing navigation accuracy,

Overall, the results indicate that decentralized pheromone-based swarms exhibit strong robustness to moderate sensing uncertainty but are considerably more vulnerable to persistent robot failures. These findings contribute to a better understanding of how decentralized multi-agent systems respond to uncertainty and provide a foundation for future investigations into more realistic swarm robotic environments and failure models.

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